

Selenium Web Automation

Overview

Selenium is the most widely used open-source web automation framework. It provides a suite of tools including WebDriver, IDE, and Grid for automating browser interactions across all major browsers and platforms.

Key Features

- Supports Chrome, Firefox, Edge, Safari, and more.
- Language support: Python, Java, C#, Ruby, JavaScript.
- Selenium Grid for distributed parallel test execution.
- Large community with extensive documentation.
- Integrates with Pytest, TestNG, JUnit, and CI/CD pipelines.

Basic Example (Python)

```
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.get('https://example.com')
print(driver.title)
driver.find_element(By.ID, 'search').send_keys('Selenium')
driver.quit()
```

Locator Strategies

Selenium provides multiple ways to locate elements on a page. Choosing the right locator improves test stability and maintenance.

Locator Examples

```
from selenium.webdriver.common.by import By

# By ID
driver.find_element(By.ID, 'username')

# By CSS Selector
driver.find_element(By.CSS_SELECTOR, '.btn-primary')

# By XPath
driver.find_element(By.XPATH, '//button[@type="submit"]')

# By Link Text
driver.find_element(By.LINK_TEXT, 'Click Here')
```

Best Practices

- Use explicit waits (WebDriverWait) instead of time.sleep().
- Follow the Page Object Model (POM) design pattern.
- Keep tests independent and atomic.
- Use CSS selectors over XPath for better performance.